

Two Structural Reductions for Written on the Wall II, Conjecture 61

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Status. Conjecture 61 is not proved here. This note proves the diameter-at-most-three case, a star-forest certificate theorem, a one-unit deletion theorem, and a two-connected escape theorem. It then isolates five explicit open statements: any one of (PI), (DL), or (FA) would finish the conjecture, (MAXP) implies (PI), and (ESC) would finish the two-connected case of (DL). Exact computation supports every open statement on 2,592,586 graph records, but finite verification is not a proof.

Abstract

For a nontrivial connected graph G , Written on the Wall II, Conjecture 61 asserts

$$f(G) \geq \text{residue}(G) + \left\lceil \frac{\text{diam}(G)}{3} \right\rceil, \quad (\text{C61})$$

where $f(G)$ is the maximum order of an induced forest and $\text{residue}(G)$ is the Havel–Hakimi residue. We give two reductions of the conjecture.

The first asks for a distance-three packing S of order $\lceil \text{diam}(G)/3 \rceil$ whose deletion leaves an independent set of order $\text{residue}(G)$. Such a packing and independent set induce a disjoint union of stars, proving the desired bound immediately. A stronger formulation asks that the packing occur among the vertices removed by a suitable maximum-degree deletion order.

The second reduction is inductive. If v is a maximum-degree vertex and $C_1, \dots, C_k$ are the components of $G - v$, then

$$\sum_{i=1}^k \left(\text{residue}(C_i) + \left\lceil \frac{\text{diam}(C_i)}{3} \right\rceil \right) \geq \text{residue}(G) + \left\lceil \frac{\text{diam}(G)}{3} \right\rceil - 1.$$

Thus the natural component induction can lose at most one vertex. We state the exact deletion lemma that would remove this last unit and prove that it implies (C61).

Both reductions are then pushed one step further. In a two-connected graph, deleting a maximum-degree vertex avoided by some diametral pair loses nothing at all, so a two-connected counterexample to the deletion lemma must have every maximum-degree vertex inside every diametral pair, diameter congruent to 1 modulo 3, and exact equality in both the residue and diameter terms after the deletion. Complementing this, a repair statement is proved: induced forests meeting the component quotas in which no tree holds two neighbors of the deleted vertex recover the lost unit whenever they exist. All five open assertions were checked exactly on every diameter-at-least-four record

among the connected unlabeled graphs through order 9, equality-boundary extensions through order 11, and a path-of-gadgets catalogue through order 20.

1. The problem in plain language

All graphs in this note are finite, simple, and undirected. The conjecture compares three quantities. Its current Lean transcription is recorded as research open [6]. The **order** of a graph is its number of vertices.

- $f(G)$ is the largest number of vertices that can be retained while the induced subgraph on those vertices has no cycle. Every edge between retained vertices must remain; edges cannot be pruned separately.
- $\text{diam}(G)$ is the greatest distance between two vertices.
- $\text{residue}(G)$ is obtained from the degree sequence by the Havel–Hakimi process: repeatedly remove the largest degree, subtract one from that many following degrees, and sort again. When only zeros remain, their number is the residue.

The residue is a classical lower bound on the independence number $\alpha(G)$, so it already certifies an induced forest of $\text{residue}(G)$ vertices. Conjecture 61 says that a connected graph also allows roughly one additional forest vertex for every three steps of diameter.

Put

$$r(G) = \text{residue}(G), \quad d(G) = \text{diam}(G), \quad q(G) = \left\lceil \frac{d(G)}{3} \right\rceil$$

and

$$b(G) = r(G) + q(G), \quad \Phi(G) = b(G) - f(G).$$

The conjecture is the assertion $\Phi(G) \leq 0$ for every nontrivial connected graph. A single graph with $\Phi(G) > 0$ would disprove it. Graphs with $\Phi(G) = 0$ are the equality boundary.

Complete graphs and stars show that equality is common for unrelated reasons. If $n \geq 2$, then K_n has $r(K_n) = 1$, $d(K_n) = 1$, and $f(K_n) = 2$. The star $K_{1,n-1}$ has residue $n - 1$, diameter 2, and is itself a forest. Both have $\Phi = 0$.

Long paths with dense decorations are the natural pressure point. The long spine raises the diameter term, while cliques, odd cycles, and dense blocks attached along the spine try to make every large induced subgraph cyclic. The computational constructions below were designed to push both effects at once.

2. A star-forest certificate

A **2-packing** is a set $S \subseteq V(G)$ such that every two distinct vertices of S are at distance at least 3 in G .

Proposition 1 (packing–independence certificate)

Suppose G contains a 2-packing S and an independent set $I \subseteq V(G) \setminus S$. Then $G[I \cup S]$ is a disjoint union of stars and isolated vertices. In particular,

$$f(G) \geq |I| + |S|.$$

Proof

There are no edges inside I , since I is independent, and no edges inside S , since distinct vertices of a 2-packing are not adjacent. Moreover, each vertex of I has at most one neighbor in S . Two such neighbors would be joined by a path of length two through that vertex, contrary to the definition of a 2-packing.

Every nontrivial component of $G[I \cup S]$ therefore consists of one vertex of S and some of its neighbors in I . It is a star. The remaining vertices are isolated, so the entire induced subgraph is a forest. $\square$

Proposition 1 turns Conjecture 61 into the following concrete structural question.

Packing–independence lemma (PI, open)

Every nontrivial connected graph G has a 2-packing S such that

$$|S| = q(G) \quad \text{and} \quad \alpha(G - S) \geq r(G). \quad (\text{PI})$$

Corollary 2

If (PI) holds for every nontrivial connected graph, then Conjecture 61 holds.

Proof

Choose an independent set I of order $r(G)$ in $G - S$, then apply Proposition 1:

$$f(G) \geq |I| + |S| = r(G) + q(G) = b(G). \quad \square$$

This reduction explains the divisor 3. Vertices chosen three steps apart cannot acquire a common neighbor in the independent part, so the combined induced graph has no cycle. Merely choosing mutually nonadjacent vertices would not be enough.

3. A maximum-degree formulation

The classical **MAX** algorithm repeatedly deletes a vertex of maximum degree until the remaining graph is independent. When several vertices have maximum degree, the result can depend on the tie-breaking.

Griggs and Kleitman proved that every possible independent set left by **MAX** has order at least $r(G)$ [2]. If **MAX** stops with more than $r(G)$ vertices, we may continue deleting isolated vertices until exactly $r(G)$ remain; a zero-degree vertex is then still a maximum-degree vertex.

This gives a proof-shaped strengthening of (PI).

MAX-packing lemma (MAXP, open)

For every nontrivial connected graph G , the maximum-degree ties can be broken so that, after exactly $|V(G)| - r(G)$ deletions, the deleted set contains a 2-packing of order $q(G)$, with distances measured in the original graph G .

Proposition 3

$$(\text{MAXP}) \implies (\text{PI}) \implies (\text{C61}).$$

Proof

Run the ordering promised by (MAXP), and let D be the first $|V(G)| - r(G)$ deleted vertices. Its complement I is independent and has order $r(G)$. If $S \subseteq D$ is the promised 2-packing, then $I \subseteq V(G) \setminus S$, so $\alpha(G - S) \geq r(G)$. This is (PI), and Corollary 2 finishes the implication. $\square$

The existential tie-breaking clause cannot simply be dropped. Among the 21,332 connected unlabeled graphs of order 9 and diameter at least 4, all admit at least one successful maximum-degree ordering, but only 10,604 work for every ordering.

There is also a technical trap in any recursive proof of (MAXP): deletion can increase distances. A set that is a 2-packing in a later residual graph need not have been a 2-packing in the original graph. An induction must therefore remember original distance-two conflicts, not only the current graph.

4. What can already be proved

Proposition 4 (one vertex beyond independence)

Every nontrivial connected graph satisfies

$$f(G) \geq \alpha(G) + 1.$$

Proof

Let A be a maximum independent set. Since G is connected and has at least two vertices, $A \neq V(G)$; choose $v \notin A$. The graph induced by $A \cup \{v\}$ is a star centered at v , together with isolated vertices. It is an induced forest of order $\alpha(G) + 1$. $\square$

Corollary 5 (diameter at most three)

Conjecture 61 holds whenever $d(G) \leq 3$.

Proof

A nontrivial connected graph has $1 \leq d(G) \leq 3$, hence $q(G) = 1$. The classical residue bound $r(G) \leq \alpha(G)$ [1] and Proposition 4 give

$$f(G) \geq \alpha(G) + 1 \geq r(G) + q(G). \quad \square$$

The only unresolved case therefore has diameter at least 4.

5. The one-unit deletion theorem

For a graph H with connected components $C_1, \dots, C_k$, define the component bound

$$b^\oplus(H) = \sum_{i=1}^k (r(C_i) + q(C_i)).$$

For an isolated vertex K_1 , we use $r(K_1) = 1$, $d(K_1) = 0$, and $q(K_1) = 0$, so its contribution to $b^\oplus$ is 1.

Lemma 6 (diameter loss under one deletion)

If G is connected, $v \in V(G)$, and $C_1, \dots, C_k$ are the components of $G - v$, then

$$\sum_{i=1}^k q(C_i) \geq q(G) - 1. \tag{1}$$

Proof

Let

$$P = x_0 x_1 \cdots x_d$$

be a diametral geodesic of G , so $d = d(G)$. If $v \notin V(P)$, the whole path survives in one component of $G - v$. Distances cannot decrease when a vertex is deleted, so that component has diameter at least d , and the left side of (1) is at least $q(G)$.

Now suppose $v = x_i$. If neither endpoint was deleted and x_0, x_d lie in one component of $G - v$, their distance there is at least d , and the same argument applies. Otherwise, the surviving arm or arms of $P - v$ lie in their respective components. Let their edge-lengths be $a, b \geq 0$, taking the missing arm to have length 0 when an endpoint of P is deleted. Each nonempty arm remains a geodesic in its component, and

$$d \leq a + b + 2.$$

Consequently,

$$\left\lceil \frac{a}{3} \right\rceil + \left\lceil \frac{b}{3} \right\rceil \geq \left\lceil \frac{a+b+2}{3} \right\rceil - 1 \geq \left\lceil \frac{d}{3} \right\rceil - 1.$$

For the first inequality, use $a \leq 3\lceil a/3 \rceil$ and $b \leq 3\lceil b/3 \rceil$. Each existing arm lies in a component whose diameter is at least its length; a missing arm contributes 0. This proves (1). $\square$

Lemma 7 (residue under a maximum-degree deletion)

Let v be a maximum-degree vertex of G , and let $C_1, \dots, C_k$ be the components of $G - v$. Then

$$\sum_{i=1}^k r(C_i) \geq r(G). \quad (2)$$

Proof

Let π' be the Havel–Hakimi reduction of the degree sequence of G . It removes the maximum degree and subtracts one from the largest available terms. Deleting the actual vertex v instead subtracts one from the terms corresponding to its neighbors. After sorting, this actual residual degree sequence dominates π' , meaning its initial partial sums are at least as large: subtracting from any terms other than the largest ones can only make the sequence more concentrated. Residue is monotone under this dominance order [1], so

$$r(G - v) \geq r(\pi') = r(G).$$

The residue of a disjoint union is at most the sum of the residues of its components [3]. Therefore

$$r(G) \leq r(G - v) \leq \sum_{i=1}^k r(C_i). \quad \square$$

Theorem 8 (one-unit deletion theorem)

If v is a maximum-degree vertex of a connected graph G , then

$$b^\oplus(G - v) \geq b(G) - 1. \quad (3)$$

Proof

Add inequalities (1) and (2). $\square$

The theorem says that the natural induction is never badly wrong. A maximum-degree deletion preserves the full residue contribution and can lose at most one unit from the rounded diameter contribution.

6. The exact inductive finishing lemma

The one missing unit can be isolated as a second open statement.

Deletion lemma (DL, open)

Every connected graph G with $d(G) \geq 4$ has a vertex v such that

$$b^\oplus(G - v) \geq b(G). \quad (\text{DL})$$

The vertex in (DL) is not required to have maximum degree, and $G - v$ may be disconnected.

Theorem 9

If (DL) holds, then Conjecture 61 holds.

Proof

Proceed by strong induction on $|V(G)|$. The case $d(G) \leq 3$ is Corollary 5. Otherwise choose v as in (DL), and let $C_1, \dots, C_k$ be the components of $G - v$.

Each nontrivial component has fewer vertices than G , so by induction it contains an induced forest of order at least $b(C_i)$. An isolated component contributes its single vertex. The union of these forests is an induced forest in $G - v$, and hence in G , of order at least

$$\sum_{i=1}^k b(C_i) = b^\oplus(G - v) \geq b(G).$$

This is (C61). $\square$

Theorem 8 sharply constrains a smallest counterexample to Conjecture 61. If G were such a graph, component induction would give $b^\oplus(G - v) \leq b(G) - 1$ for every vertex v . When v has maximum degree, integrality and (3) would therefore force

$$b^\oplus(G - v) = b(G) - 1$$

for every such v . Both ingredients would have to be tight:

$$\sum_i r(C_i) = r(G), \quad \sum_i q(C_i) = q(G) - 1. \quad (4)$$

This reduces the next proof attempt to the equality structure in (4): show that either another vertex satisfies (DL) or an induced forest through the deleted vertex recovers the missing unit. The next two sections carry both alternatives as far as they currently go.

7. The two-connected case

In the equality structure (4), the diameter term loses its unit only under a genuine separation. Making this precise closes the two-connected case up to a single structural question.

Lemma 10 (metric loss requires separation)

Let v be a vertex of a connected graph G , and suppose some diametral pair $\{x, y\}$ of G satisfies $v \notin \{x, y\}$, with x and y in the same component C of $G - v$. Then

$$\sum_{i=1}^k q(C_i) \geq q(G).$$

Proof

Deleting a vertex cannot decrease distances, so the distance between x and y inside C is at least $d(G)$. Hence $d(C) \geq d(G)$ and $q(C) \geq q(G)$. $\square$

This is the first case in the proof of Lemma 6, restated because its contrapositive is what matters here: losing the diameter unit forces the deleted vertex to separate, or to belong to, every diametral pair. No vertex separates a two-connected graph.

Theorem 11 (two-connected escape)

Let G be two-connected with $d(G) \geq 4$, and let v be a maximum-degree vertex avoided by some diametral pair. Then $G - v$ is connected and

$$b(G - v) \geq b(G).$$

In particular, v satisfies (DL).

Proof

Since G is two-connected, $G - v$ is connected, so the avoided diametral pair lies in the single component $G - v$ and Lemma 10 gives $q(G - v) \geq q(G)$. Lemma 7 gives $r(G - v) \geq r(G)$. Adding the two inequalities proves the claim. $\square$

Corollary 12 (structure of a trapped counterexample)

Suppose G is two-connected with $d(G) \geq 4$ and no vertex of G satisfies (DL). Then:

1. every maximum-degree vertex lies in every diametral pair; consequently G has at most two maximum-degree vertices, and if it has exactly two, the diametral pair is unique;
2. $d(G) \equiv 1 \pmod{3}$;
3. for every maximum-degree vertex v , both $d(G - v) = d(G) - 1$ and $r(G - v) = r(G)$ hold exactly;
4. for every diametral pair $\{v, y\}$ with v of maximum degree, v is the only vertex of G at distance $d(G)$ from y .

Proof

Claim 1 is Theorem 11 read contrapositively; a diametral pair has two elements, and two distinct maximum-degree vertices can both lie in every pair only if every pair is exactly those two vertices.

For claim 4, if some $z \neq v$ had $\text{dist}(z, y) = d(G)$, then $\{z, y\}$ would be a diametral pair avoiding v , against claim 1.

For claims 2 and 3, fix a maximum-degree v and a diametral pair $\{v, y\}$. The graph $G - v$ is connected and retains the arm of a diametral geodesic from y , so $d(G - v) \geq d(G) - 1$, while Lemma 7 gives $r(G - v) \geq r(G)$. If $d(G) \not\equiv 1 \pmod{3}$, then $\lceil (d(G) - 1)/3 \rceil = q(G)$ and $b(G - v) \geq b(G)$, contradicting the absence of a (DL) vertex. So $d(G) \equiv 1 \pmod{3}$, and $b(G - v) \leq b(G) - 1$ forces equality throughout: $r(G - v) = r(G)$ and $q(G - v) = q(G) - 1$. Since $d(G) \equiv 1 \pmod{3}$ gives $3(q(G) - 1) = d(G) - 1$, the ceiling equality caps $d(G - v) \leq d(G) - 1$, so $d(G - v) = d(G) - 1$ exactly. $\square$

The remaining question is whether trapped graphs exist at all.

Escape lemma (ESC, open)

Every two-connected graph with $d(G) \geq 4$ has a maximum-degree vertex avoided by some diametral pair.

By Theorem 11, (ESC) implies (DL) for every two-connected graph. All 53,096 two-connected diameter-at-least-four records in the four corpora satisfy (ESC); no trapped graph has been seen (Section 9.4). Two cautions attach to that evidence. First, coverage: in a two-connected graph every intermediate breadth-first level from any vertex has at least two vertices, since a singleton level would be a cut vertex separating what lies beyond it; hence $|V(G)| \geq 2d(G)$, and the exhaustive order-at-most-9 slice tests (ESC) only at $d(G) = 4$. Evidence at larger diameters comes from the structured families alone. Second, a trapped graph would not by itself refute (DL) or Conjecture 61: Corollary 12 would still leave its rigid residual structure open to a direct argument.

8. Forests through the deleted vertex

The second alternative in the equality analysis of (4) keeps the deleted vertex in the forest.

Proposition 13 (free-attachment repair)

Let v be a maximum-degree vertex of a connected graph G with $d(G) \geq 4$, and let $C_1, \dots, C_k$ be the components of $G - v$. Suppose each C_i contains an induced forest F_i with $|F_i| \geq b(C_i)$ such that no tree of $C_i[F_i]$ contains two neighbors of v . Then

$$f(G) \geq b(G).$$

Proof

The union $F = F_1 \cup \dots \cup F_k$ is an induced forest of $G - v$. Any cycle in $G[F \cup \{v\}]$ would have to pass through v , and its remainder would be a path in the forest $G[F]$ joining two neighbors of v . A path in a forest stays inside one tree, so some tree would contain two neighbors of v , contrary to hypothesis. Adding v merges the trees it touches into one and creates no cycle, so $G[F \cup \{v\}]$ is an induced forest of order at least

$$1 + \sum_{i=1}^k b(C_i) = 1 + b^\oplus(G - v) \geq b(G)$$

by Theorem 8. $\square$

Free-attachment lemma (FA, open)

Every connected graph G with $d(G) \geq 4$ has a maximum-degree vertex v admitting forests as in Proposition 13.

Proposition 14

If (FA) holds, then Conjecture 61 holds.

Proof

Corollary 5 covers $d(G) \leq 3$; Proposition 13 covers $d(G) \geq 4$ directly. $\square$

The hypotheses of (FA) cannot simply be dropped. If a component is a single edge with both endpoints adjacent to v , the only forest meeting the quota $b(K_2) = 2$ is the edge itself, and its single tree holds both neighbors. That configuration is a triangle and has diameter 1; the open content of (FA) is that diameter at least 4 and maximum degree leave enough room. The star-forest witness of Section 2 is the natural candidate for the F_i : stars are hard to enter twice, which is one more sign that the packing reduction and the deletion reduction aim at the same object.

At every maximum-degree deletion in the corpora that is one-unit tight — $b^\oplus(G - v) = b(G) - 1$, the only case Theorem 8 leaves open, since otherwise v itself satisfies (DL) — the constrained quota forests exist: 58,809 configurations, no exceptions (Section 9.4).

9. Exact computational evidence

The computation had two roles: search directly for $\Phi(G) > 0$, and test the proposed structural lemmas without replacing them by heuristics.

9.1 Evaluator validation

The optimized C evaluator was compared with separate exhaustive oracles on all 33,867 labeled graphs through order 6:

- manual Havel–Hakimi reduction against the production residue routine;
- Floyd–Warshall against repeated-BFS diameter;
- all vertex subsets against the cycle-branching induced-forest optimizer;
- graph6 encoding and decoding round trips.

Complete graphs, paths, and stars were also checked through order 20. An independent Python program re-evaluated reported boundary graphs by descending subset enumeration.

9.2 Counterexample search

Four complementary searches were used.

1. **Exhaustive small graphs.** Every connected unlabeled graph through order 9 was generated. The order-9 count of 261,080 agrees with McKay’s published table [4].
2. **Path-of-gadgets constructions.** The catalogue joins cliques, odd cycles, complete bipartite blocks, barbells, articulation blocks, mixed blocks, and layered paths along thin necks, through order 20.
3. **Boundary extensions.** Every nonempty neighborhood for a new vertex was attached to each of the 1,231 order-9 equality graphs. Leaves and true or false twins were then attached to the 49,521 surviving order-10 equality records.
4. **Direct optimization.** Edge additions and deletions preserving connectivity hill-climbed Φ and the smoother score $3(r(G) - f(G)) + d(G)$. About 7.5 million edge toggles were considered at orders 10, 12, 13, 16, and 18, with every connected candidate evaluated exactly.

No graph with $\Phi(G) > 0$ was found. This is a bounded null result, not a proof of (C61).

9.3 Structural-lemma checks

[proof61.c](#) enumerates 2-packings and solves the remaining independent-set decision problem exactly. [maxine61.c](#) uses dynamic programming over remaining-vertex masks, explores every allowed maximum-degree tie, and inspects the frontier at exactly $r(G)$ remaining vertices.

corpus	records	$d \leq 3$	$d \geq 4$	(PI)	some MAX order	every MAX order
connected unla- beled graphs of order 9	261,080	239,748	21,332	21,332	21,332	10,604
all one- vertex exten- sions of order-9 equality graphs	629,041	509,354	119,687	119,687	119,687	37,579
leaf/twin exten- sions of surviving order-10 equality records	1,342,198	962,218	379,980	379,980	379,980	111,752
path-of- gadgets cata- logue through order 20	360,267	376	359,891	359,891	359,891	213,501
total evalu- ated records	2,592,586	1,711,696	880,890	880,890	880,890	373,436

The extension and family rows contain isomorphic duplicates. The totals therefore count exact evaluations, not distinct isomorphism classes. A separate component-bound scan found a vertex satisfying (DL) in all 880,890 records of diameter at least 4.

The deletion reduction gives a more detailed picture at order 9. Of the 21,332 graphs not covered by Corollary 5:

- 15,568 have a successful maximum-degree deletion that leaves a connected graph;
- 21,136 have some successful connected deletion;

- the remaining 196 have no successful connected deletion, but all 196 have a cut-vertex deletion satisfying (DL).

Thus no order-9 graph is induction-critical. Thirty-one of the final 196 graphs also lie on the equality boundary $\Phi = 0$.

9.4 Endgame checks

[finish61.c](#) ran three further exact checks across the same four corpora. Its internal deletion census independently reproduced the order-9 numbers above.

Lemma 7 stress test. All 5,246,681 maximum-degree deletions across the 2,592,586 records satisfied $\sum_i r(C_i) \geq r(G)$, with zero violations. This exercises the dominance step of Lemma 7 — the one step of the proved material resting on a monotonicity property cited from [1] — on every available record.

(ESC) and (FA) census. Every two-connected record of diameter at least 4 has a maximum-degree vertex avoided by some diametral pair, and the conclusion $b(G - v) \geq b(G)$ of Theorem 11 was re-verified by direct computation at each such vertex, with no failures. Every one-unit-tight maximum-degree deletion admits the constrained quota forests of Proposition 13; the per-component search was exhaustive over vertex subsets, with no records skipped.

corpus	2-connected $d \geq 4$	trapped	tight deletions	repaired
connected unlabeled graphs of order 9	1,944	0	854	854
all one-vertex extensions of order-9 equality graphs	18,037	0	7,113	7,113
leaf/twin extensions of surviving order-10 equality records	30,380	0	28,936	28,936
path-of- gadgets catalogue through order 20	2,735	0	21,906	21,906
total	53,096	0	58,809	58,809

10. False strengthenings

Exact data rules out three tempting shortcuts.

1. Replacing residue by independence is false:

$$f(G) \geq \alpha(G) + q(G)$$

fails on 539 connected graphs of order 9.

2. Letting $\rho_2(G)$ be the maximum order of a 2-packing, the stronger inequality

$$f(G) \geq r(G) + \rho_2(G)$$

fails on 114 order-9 graphs with diameter at least 4. A maximum packing need not leave enough room for the independent part.

3. Replacing $\alpha(G - S) \geq r(G)$ in (PI) by $r(G - S) \geq r(G)$ is too strong. Ten connected order-9 graphs have no target-size packing satisfying the residue condition, although each has a packing satisfying the independence condition in (PI).

These failures locate the useful level of generality. The target-size packing and the independent set left after its deletion survive; the more obvious stronger statements do not.

11. What remains

There are now three independent finishing routes,

$$(\text{MAXP}) \implies (\text{PI}) \implies (\text{C61}), \quad (\text{DL}) \implies (\text{C61}), \quad (\text{FA}) \implies (\text{C61}),$$

and one partial route: by Theorem 11, (ESC) implies the two-connected case of (DL).

The two-connected case is the most constrained. Corollary 12 pins any counterexample there to diameter congruent to 1 modulo 3, at most two maximum-degree vertices locked inside every diametral pair, and exact equality in both deletion terms. Settling (ESC) is the natural next step. A trapped graph, if one exists, has at least $2d(G) \geq 14$ vertices when $d(G) \geq 7$, so it lies beyond the exhaustive corpus; targeted constructions — a high-degree fan feeding a long ladder, at diameters 7 and 10 — would either produce one or sharpen the evidence. For $d(G) = 4$ the exhaustive data covers only orders 8 and 9.

On the block-tree side, (FA) is the statement that still lacks a mechanism. The 58,809 successful repairs say the constrained forests always exist in the tested range; they do not say why. The mod-3 arithmetic of Lemma 6 favors cut vertices at positions congruent to 1 modulo 3 along a diametral geodesic — the same spacing as the packing skeleton of Section 2 — but such a vertex must also preserve the residue sum, and residue is unprotected for deletions below maximum degree.

A counterexample to (DL), (PI), (MAXP), (ESC), or (FA) would not necessarily be a counterexample to Conjecture 61; it would only rule out that proof route. Further unconstrained edge mutation is unlikely to resolve the question; the equality classification that Section 6 called for has now been carried into Sections 7 and 8, and what remains of it is exactly (ESC) and (FA).

The related WOWII Conjecture 17 says that the maximum order of an induced bipartite subgraph is at least $\alpha(G) + \lceil d(G)/3 \rceil$; it is listed as resolved on the historical WOWII page [5]. Its diameter

term has the same spacing geometry, but bipartiteness permits even cycles. Proposition 1 uses distance three to obtain the stronger conclusion that the witness is a forest.

12. Reproduction

The implementation is in this directory. All build products are written to /tmp.

```
cd problems/wow2
make test
make check-counts-9
make exhaustive-61
make families-61
make check-critical-61
make check-maxine-61
make check-finish-61
```

The direct order-9 packing scan is:

```
/tmp/math-eval-wow2/wow2-generate 9 \
| /tmp/math-eval-wow2/wow2-proof61 --scan-packing
```

The exact evaluator and graph routines are in `graph.c` and `conjectures.c`. The endgame checks of Section 9.4 are implemented in `finish61.c`; any graph6 stream can be piped through the resulting `wow2-finish61`. The independent verifier is `verify.py`. The complete bounded-search log, including graph6 records on the equality boundary, is in `RESULT-061.md`.

References

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